

Networking Fundamentals

Unit 03 — Technical Foundations

You can't attack — or defend — a network you don't understand. This week we learn how data actually travels, from zero.

Learning objectives

By the end of this unit you can:

- **Describe**, in order, how data travels from one device to another.
- **Distinguish** an IP address from a MAC address.
- **Identify** an IPv4 address as public or private.
- **Match** 8+ common ports to their services.
- **Explain** TCP vs. UDP and give a use of each.
- **Diagram** the TCP three-way handshake (SYN → SYN-ACK → ACK).
- **Describe** the 4-layer TCP/IP model and relate it to the OSI model.
- **Trace** a DNS lookup from a name to an IP.
- **Capture** live traffic in Wireshark and identify protocol, IPs, and ports.

What is a network?

- A **network** = two or more devices connected to share data.
- A **host** = any device on it (computer, phone, printer, server).
- A **router** forwards packets between different networks.
- A **packet** = a small chunk of data wrapped with addressing info.

Big picture: data is chopped into packets, each labeled with where it's from and where it's going, then forwarded hop by hop.

IP vs. MAC address

	IP address	MAC address
Looks like	192.168.1.10	00:1A:2B:3C:4D:5E
What it is	Logical address — <i>where</i> you are	Hardware ID burned into the card
Changes?	Yes (it's where you are now)	No (fixed to the device)
Used for	Routing across the internet	Delivery on the local segment

Public vs. private IPs

These three blocks are **private** — used inside a local network, not directly reachable from the internet:

- `10.0.0.0/8` → `10.x.x.x`
- `172.16.0.0/12` → `172.16.x` through `172.31.x`
- `192.168.0.0/16` → `192.168.x.x`

Everything else routable is **public**. Loopback is `127.0.0.1`.

Ports & services

One host (one IP) can run many services — each listens on a numbered **port** (like apartment doors at one street address).

Port	Service	Port	Service
20/21	FTP	80	HTTP
22	SSH	443	HTTPS
23	Telnet	445	SMB
25	SMTP	3389	RDP
53	DNS		

TCP/IP & OSI models

TCP/IP (4 layers)	Example	OSI (rough)
Application	HTTP, DNS, SSH	5-7
Transport	TCP, UDP (ports)	4
Internet	IP (IP addresses)	3
Link	Ethernet (MAC)	1-2

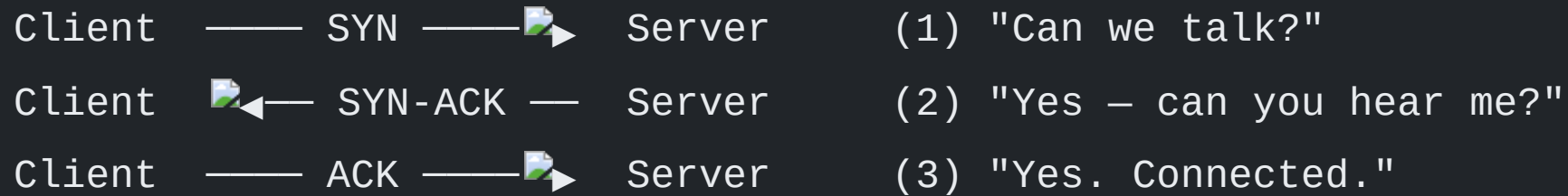
Don't memorize all 7 OSI layers — just answer "which layer is this thing on?" The 4-layer model is what you'll actually use.

TCP vs. UDP

	TCP	UDP
Connection?	Yes — confirms delivery	No — just sends
Speed	Slower, reliable, ordered	Fast, no guarantee
Use it for	Web, SSH, file transfer	DNS lookups, video/voice streaming

Analogy: TCP is a **phone call** (you confirm the other side hears you). UDP is **dropping a postcard** in the mail.

The TCP three-way handshake



- **SYN-ACK comes back from the server** — it's **one** packet, not two SYNs.
- After ACK, the connection is established and data flows.

DNS: the internet's phonebook

1. You type `example.com` — your computer doesn't know its IP yet.
2. **Stub resolver** asks a **recursive resolver**.
3. Resolver walks **root → TLD → authoritative** servers.
4. An **A record** (the IPv4 address) comes back.
5. Your browser connects to that IP.

DNS queries usually ride **UDP port 53**. A query asks "what's the IP for this name?"; the response carries the answer.

Ethics & Authorization

Packet capture is wiretapping when the traffic isn't yours.

Capturing on a network you **own or are authorized to test** (this classroom lab, your own home network) is fine. Capturing **other people's** traffic — public Wi-Fi, a friend's network, the school network outside this lab — can break wiretap and computer-crime laws **even if you "just looked."**

The dividing line, as always, is **authorization and scope.**

Reading packets: Wireshark & tcpdump

- **Wireshark** = a graphical tool to capture and inspect packets.
- **tcpdump** = the command-line version.
- **Display filters** let you focus the view:

Filter	Shows
<code>icmp</code>	Ping (echo request/reply) packets
<code>dns</code>	DNS queries and responses
<code>tcp.port == 80</code>	HTTP traffic
<code>ip.addr == <IP></code>	All traffic to/from one host

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Key vocabulary

Term	Meaning
IP / MAC address	Logical (changeable) address / fixed hardware ID
Public / Private IP	Reachable on the internet / local-network only
Port / Service	A numbered door / the program answering on it
Protocol	Agreed rules for two devices talking (HTTP, DNS, TCP)
TCP / UDP	Reliable & connection-based / fast & connectionless
Three-way handshake	SYN → SYN-ACK → ACK to start a TCP connection
DNS	Turns names like <code>example.com</code> into IP addresses
Packet / Wireshark / tcpdump	Chunk of data / GUI capture tool / CLI capture tool

Lab launch

Platform: TryHackMe networking room (browser, free tier) **and** Wireshark on your lab machine.

- Finish the **TryHackMe networking room** to lock in concepts.
- Find your own **IP + MAC** (`ip addr` / `ipconfig /all`); label public or private.
- **Capture** a ping (`icmp`), then a web page load + DNS lookup.
- **Locate the three-way handshake** (SYN → SYN-ACK → ACK) and annotate 3 packets.

→ Full walkthrough in this unit's `lab.md`. Capture **only** on the interface your teacher authorizes.

Recap

- Data travels as **packets**; **IP** routes them, **MAC** delivers locally.
- Three **private** blocks: `10`, `172.16-31`, `192.168`.
- **Ports** map to services; **TCP** is reliable, **UDP** is fast.
- TCP starts with **SYN → SYN-ACK → ACK**.
- **DNS** turns names into IPs.
- Capturing others' traffic without authorization is **wiretapping**.

Exit ticket & discussion

Exit ticket: Put `SYN`, `ACK`, and `SYN-ACK` in the correct order — and say which side sends each.

Discuss: Where exactly is the line between "I was just curious and looked at the packets" and "I intercepted someone's communications"? What makes the classroom lab capture legal when a coffee-shop capture isn't?